

HABIB IBRAHIM TOURE

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Computer Science Co-op student at uOttawa — building full-stack and AI-powered systems

EDUCATION

University of Ottawa

Ottawa, ON

B.Sc. (Specialized) in Computer Science, Co-op | International Student Merit Scholarship – \$36,000/yr Sept. 2024 – 2028

- Relevant coursework: Data Structures and Algorithms, Computer Architecture, Introduction to Software Engineering

EXPERIENCE

Data Annotation & AI Training Specialist

June 2025 – Oct. 2025

DataAnnotation

Remote, ON

- Annotated and evaluated datasets across 5+ ML projects to train and benchmark machine learning models
- Built Python automation scripts to eliminate repetitive annotation steps, reducing manual effort by ~30%
- Analyzed model outputs and delivered structured feedback that improved labeling consistency across projects
- Identified systematic labeling errors in collaboration with a remote AI team and helped refine annotation guidelines

PROJECTS

MARA | *Python, Claude API, Whisper, PyQt5, Selenium, CUDA, Fish Audio*

Apr. 2026 – Jun. 2026

- Built a fully local voice assistant with push-to-talk trigger, achieving <1s response time on simple queries via parallel Haiku/Sonnet inference
- Designed hybrid inference pipeline: Whisper (local GPU) for STT, Claude Haiku for intent classification (~150ms), and Claude Sonnet for reasoning — all running in parallel to minimize perceived latency
- Implemented 10+ system control actions (volume, brightness, WiFi, screenshots, app management) and full browser automation via Selenium with encrypted credential storage
- Engineered persistent encrypted memory with Fernet and automatic conversation summarization to maintain context across sessions

Fretify | *Python, FastAPI, React, TypeScript, Librosa, Basic-Pitch, Demucs*

Oct. 2025 – Present

- Building an AI application that converts audio/video recordings into guitar tablature using machine learning
- Implemented audio processing pipeline with Librosa and Basic-Pitch for note detection and MIDI conversion
- Integrated Demucs source separation model to isolate guitar tracks from mixed audio, reducing note detection noise
- Designed full-stack app with FastAPI backend and React/TypeScript frontend featuring interactive tab visualization

Seren | *React, TypeScript, FastAPI, Claude API, PostgreSQL, Chrome Extension*

Apr. 2026 – Present

- Building a Chrome extension (Manifest V3) that surfaces a floating AI toolbar on any text selection — delivering instant explanations, summaries, and quiz generation via Claude API
- Architected full-stack platform with FastAPI backend, PostgreSQL database, JWT auth, and a React dashboard for deadline tracking and .ics calendar import
- Implemented SOS mode and Pomodoro focus timer in the extension popup to reduce student cognitive load during high-stress periods
- Connected extension and web platform via shared JWT session — login once on the site, state syncs instantly to the popup

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks & Libraries: React, FastAPI, Node.js, SQLAlchemy, Scikit-learn, NumPy, Librosa, Selenium, PyQt5

Databases: MySQL, SQLite, PostgreSQL

Developer Tools: Git, Docker, Linux, Chrome Extension (MV3), JWT, CI/CD, Unit Testing

Other: Bilingual (English/French), Machine Learning, API Development, CUDA / GPU Inference